

BANARAS HINDU UNIVERSITY

B.Sc. Semester II Mid-Term Test, 2024-25 (NEP)

MATHEMATICS (SEC)

Paper Code: MAT-SEC-21 / MATSE21 : $\text{\LaTeX}$ Typesetting for Beginners

Time: 01 Hour

Maximum Marks: 20

Instructions: Attempt any 03 questions including Question no. 1 which is compulsory.

Question 1. Answer all parts:

[10 × 1 = 10]

- What do you mean by $\text{\LaTeX}$?
- What are the $\text{\LaTeX}$ commands for writing the Greek letters γ and ϵ ?
- Write the $\text{\LaTeX}$ command for $A \cup B$.
- Make the following equation in $\text{\LaTeX}$: $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.
- Write a simple $\text{\LaTeX}$ program to create a file containing an itemized list.
- How to create **January 20, 2025**?
- What do you mean by hypertext links? Write a command to insert hyper link in $\text{\LaTeX}$.
- Write the command to insert an image in $\text{\LaTeX}$.
- Write a difference between `\hfill` and `\vfill`.
- Write a difference between `\vdots` and `\cdots`.

Question 2. Write the input command for the following expression in $\text{\LaTeX}$: $[3 + 2 = 5]$

(a) $f(x) = \begin{cases} \frac{\beta \sin(|x|)}{|x|} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0, \end{cases} \quad \text{at } x = 0.$

(b) $A = \begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{bmatrix}$

Question 3. Find the errors in the following $\text{\LaTeX}$ source, write a corrected version and write its output: [5]

```
\documentclass{article}
\usepackage{graphicx}
\title{An easy article}
\author{A Student}
\maketitle
\begin{document}
This article aims to prove for any  $x \in \mathbf{R}$ 

$$\left[\frac{\frac{x}{2}}{\frac{1}{4}}\right] = 2x$$

\end{document}
Which has already been proven.
```

Question 4. Write a $\text{\LaTeX}$ code to produce the following output:

[5]

$$\begin{aligned} E[|X|] &= \int_x |x| f_X dx \\ &= \int_{|x| \geq a} |x| f_X dx + \int_{|x| < a} |x| f_X dx \\ &\geq \int_{|x| \geq a} |x| f_X dx \\ &\geq a \int_{|x| \geq a} f_X dx \\ &= a E[|X| \geq a] \\ \therefore E[|X| \geq a] &\leq \frac{E[|X|]}{a} \end{aligned}$$
